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Charging Infrastructure

Lecture-40

Small Signal Modelling of PSFB-II

Dr. Apurv Kumar Yadav

Department of Electrical Engineering



Recap

$$D_{loss} = \frac{4 L_{Mag} n I_o \cdot f_{sw}}{V_{in}}$$

$$V_o = n V_{in} (D - D_{loss})$$

- to achieve $\rightarrow L_{Mag} \uparrow \rightarrow D_{loss} \uparrow$
  - $V_{in} \uparrow \rightarrow D_{loss} \downarrow$; $I_o \uparrow \rightarrow D_{loss} \uparrow$; $f_{sw} \uparrow \rightarrow D_{loss} \uparrow$

- In one wants to have ZVS of both the legs, they have compromise on D_{loss}

$$D_{eff} = D - D_{loss}$$

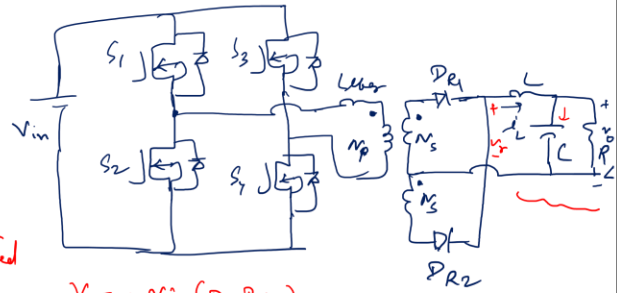
$$\tilde{d}_i = \frac{-4 n L_{Mag} f_{sw}}{V_{in}} \tilde{i}_L$$

$$\tilde{d}_v = \frac{\Delta t}{T_s/2} = \frac{n \left[2 I_o - \frac{V_o}{L} (1-D) \frac{T_s}{2} \right] \left[\frac{L_{Mag}}{V_{in}^2} \right] \tilde{V}_{in}}{T_s/2}$$

$$\tilde{d}_{eff} = \tilde{d} + \tilde{d}_i + \tilde{d}_v$$

$\Rightarrow \quad G_{vd}(s) = \dfrac{\tilde{v}_o(s)}{\tilde{d}(s)}$

$\Rightarrow$ Average large signal
↳ small perturbation
↳ linearize it around operating point
↳ Convert the model into s-domain

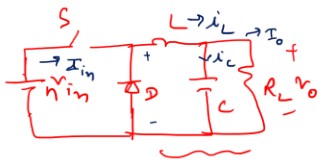


$\Rightarrow$ this is the buck derived topology

$V_o = n V_{in}(D - D_{loss})$

$T_s = T_{s1} + T_{s2}$

DT_s (S is ON)





$\Rightarrow \quad (nV_{in} - v_o) = L\dfrac{di_L}{dt} \quad \Rightarrow \quad L\dfrac{di_L}{dt} = -v_o + nV_{in}$

$\Rightarrow \quad i_C = i_L - \dfrac{v_o}{R_L} \quad \Rightarrow \quad C\dfrac{dv_c}{dt} = i_L - \dfrac{v_c}{R_L}$

$\Rightarrow \quad v_o = v_c$

$\Rightarrow \quad I_{in} = i_L$

$(1-D)T_s'$ (S is off, D is on)

$$L\frac{di_L}{dt} = -v_o$$

$$C\frac{dv_c}{dt} = i_L - \frac{v_c}{R_L}$$

$$v_o = v_c$$

$$I_{in} = 0$$

Average model (large signal model)

$$L\frac{di_L}{dt} = \frac{(-v_o + nv_{in})DT_s' - v_o(1-D)T_s'}{T_s'}$$

$$L\frac{di_L}{dt} = -v_o + nv_{in}D \longrightarrow (1)$$

$$C\frac{dv_c}{dt} = \frac{\left(i_L - \frac{v_c}{R_L}\right)DT_s' + \left(i_L - \frac{v_c}{R_L}\right)(1-D)T_s'}{T_s'}$$

$$C\frac{dv_c}{dt} = i_L - \frac{v_c}{R_L} \longrightarrow (2)$$

$$v_c = v_o \longrightarrow (3)$$

$$I_{in} = D i_L \longrightarrow (4)$$

$D,\ v_c,\ i_L,\ nv_{in}$ (Small perturbation around the operating point)

$D = D + \tilde{d}$

$v_c = v_c + \tilde{v_o}$

$i_L = I_L + \tilde{i_L}$

$nv_{in} = nv_{in} + n\tilde{v_{in}}$

$D,\ v_c,\ I_L,\ nv_{in}$ are the steady state quantities

$\tilde{d},\ \tilde{v_o},\ \tilde{i_L},\ n\tilde{v_{in}}$ are the small signal quantities

$\Rightarrow$ $D >> \tilde{d}$

$v_c >> \tilde{v_o}$

$I_L >> \tilde{i_L}$

$nv_{in} >> n\tilde{v_{in}}$

$\Rightarrow$ Multiplication of two small signal quantities $= 0$

↳ Small signal approximation

Steady state equations $\left(\frac{dI_L}{dt} = 0;\ \frac{dv_c}{dt} = 0 \right)$

$\Rightarrow$ $-v_o + nv_{in} D = 0$

$v_c = v_o$

$I_{in} = D I_L$

$\Rightarrow I_L = \frac{v_o}{R_L}$

$$L\frac{di_L}{dt} = -v_o + nv_{in}D$$

$$C\frac{dv_c}{dt} = i_L - \frac{v_c}{R_L}$$

$v_c = v_o$

$I_{in} = D\, i_L$

$$i_L = I_L + \tilde{i}_L$$

$$v_c = V_c + \tilde{v}_o \Rightarrow v_o = V_o + \tilde{v}_o$$

$$D = D + \tilde{d}$$

$$nv_{in} = nV_{in} + n\tilde{v}_{in}$$

$$I_{in} = I_{in} + \tilde{I}_{in}$$

$$L\frac{d}{dt}(I_L + \tilde{i}_L) = -(V_o + \tilde{v}_o) + (nV_{in} + n\tilde{v}_{in})(D + \tilde{d}) \longrightarrow (5)$$

$$\Rightarrow C\frac{d(V_c + \tilde{v}_o)}{dt} = (I_L + \tilde{i}_L) - \frac{V_c}{R_L} - \frac{\tilde{v}_o}{R_L} \longrightarrow (6)$$

$$\Rightarrow V_c + \tilde{v}_c = V_o + \tilde{v}_o \longrightarrow (7)$$

$$I_{in} + \tilde{I}_{in} = (D + \tilde{d})(I_L + \tilde{i}_L) \longrightarrow (8)$$

from steady state

$$\frac{dI_L}{dt} = 0 \; ; \; \frac{dv_c}{dt} = 0$$

$$+nV_{in}D - V_o = 0$$

$$I_{in} = DI_L \; ; \; V_o = V_c$$

$$I_L = \frac{V_c}{R_L}$$

$$L\frac{d\tilde{i}_L}{dt} = -\tilde{v}_o + nv_{in}\tilde{d} + n\tilde{v}_{in}D \longrightarrow (9)$$

$$C\frac{d\tilde{v}_c}{dt} = \tilde{i}_L - \frac{\tilde{v}_o}{R_L} \longrightarrow (10)$$

$$\tilde{v}_o = \tilde{v}_o \longrightarrow (11)$$

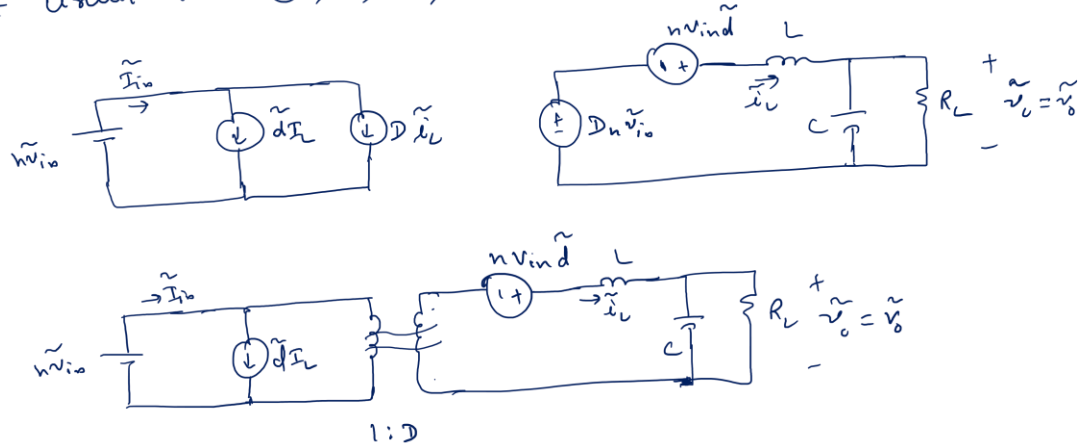
$$\tilde{I}_{in} = I_L\tilde{d} + D\tilde{i}_L \longrightarrow (12)$$

(Considering the small signal approximation

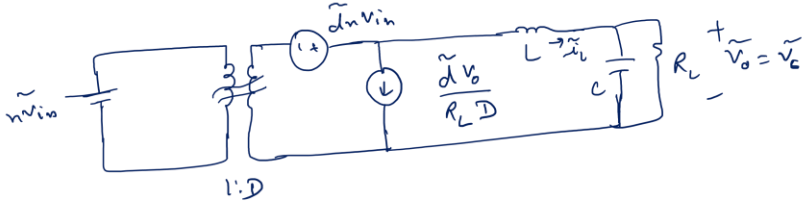
$\Rightarrow n\tilde{v}_{in}\cdot\tilde{d} = 0$

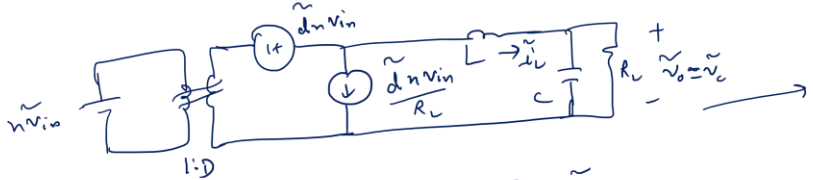
$\tilde{d}\cdot\tilde{i}_L = 0$

Equivalent circuit for (9), (10), (11), (12) are



Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.

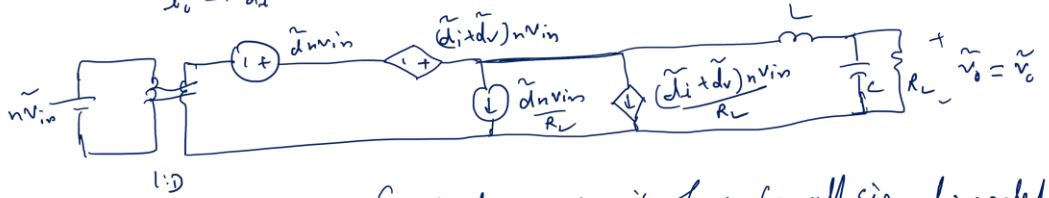




Equivalent circuit for small signal model of Buck converter

$$\tilde{d}_{eff} = \tilde{d} + \tilde{d}_i + \tilde{d}_v$$

$$\tilde{i}_o \rightarrow \tilde{d}_i \quad \Rightarrow \quad \tilde{v}_{in} \rightarrow \tilde{d}_v$$



$\Rightarrow$ Equivalent circuit for small signal model of PSFB converter.

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Thank You